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Delhi Metro Rail Corporation Limited



Safety Through Stability: A Conicity-Driven Analysis of Rail Grinding and Vehicle Stability - Field Evidence from the India's Largest Metro Network

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Introduction: India's Largest Metro Network

Delhi Metro Rail network is India's largest metro rail network.

416 km

Network length

303

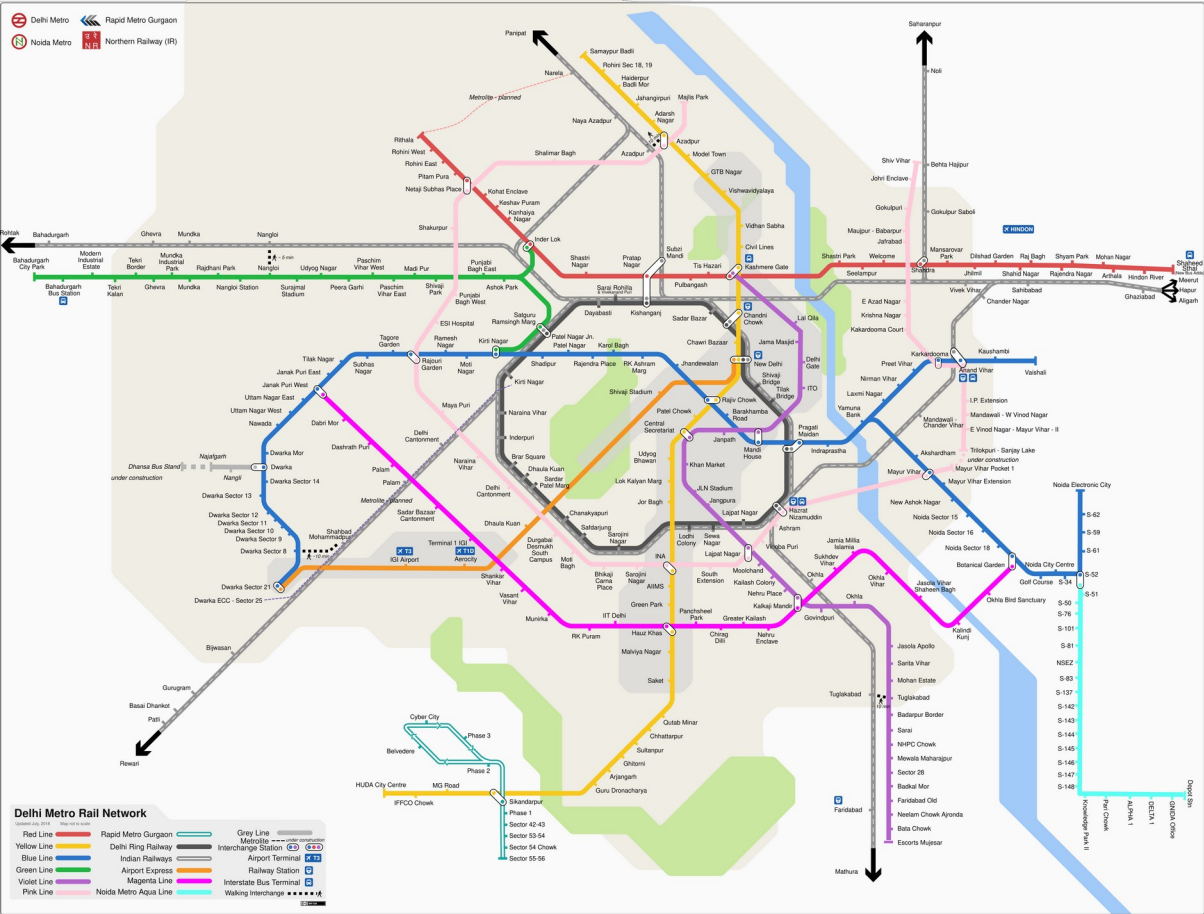
Locations connected

2.88+ Billion

Annual passengers

40+

Tangent sections studied



Introduction: Milestones & Network Expansion

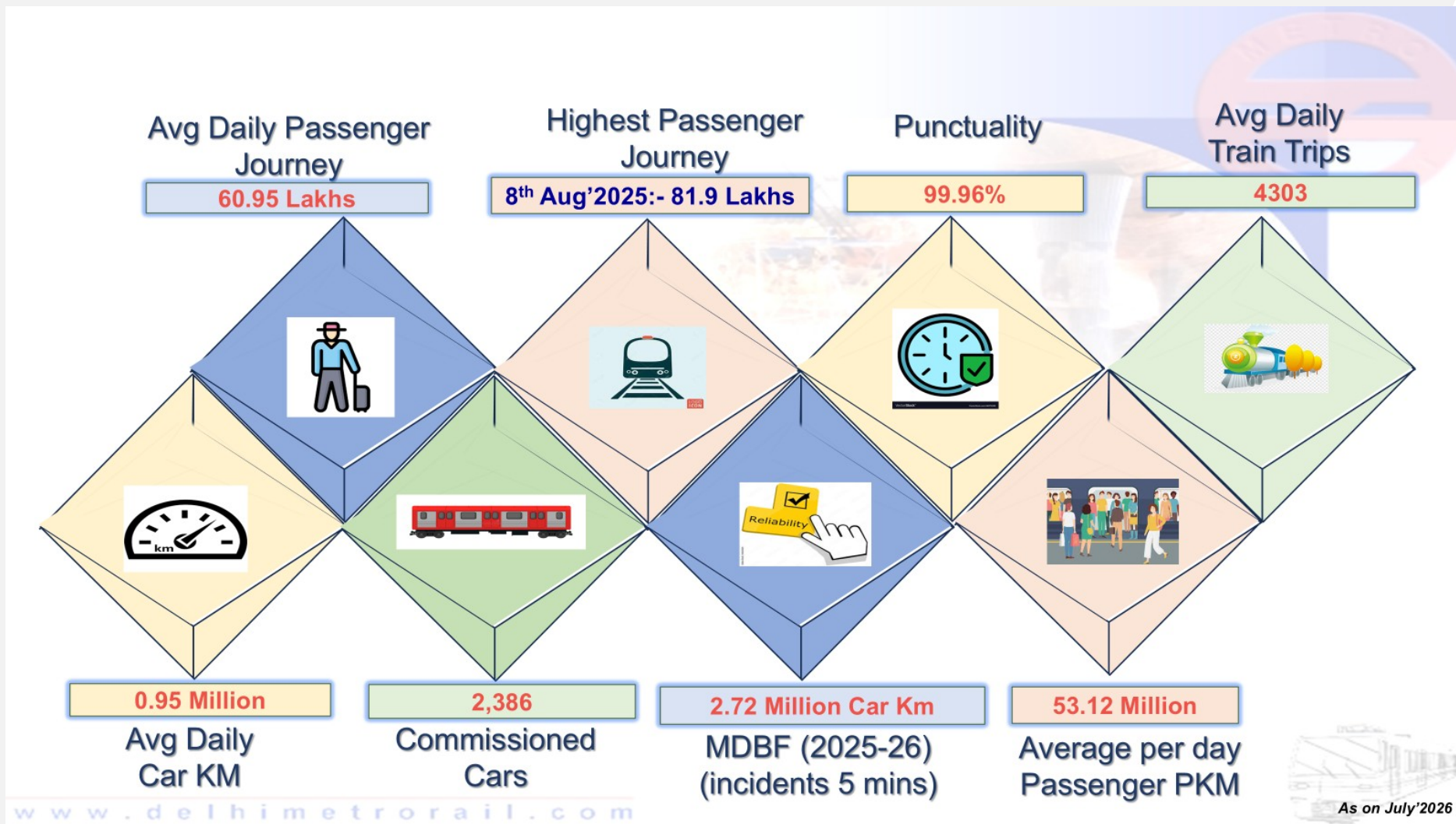


- As of now, DMRC operates **416.488 km** with **303 stations**, including **29 km** (21 stations) of the Noida-Greater Noida Line and **13 km** of Rapid Metro (11 stations) of Rapid Metro with **31 interchanges**.



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Introduction: Operational Highlights



It became the first metro system globally to be registered as a 'Carbon Credit' project by the UN, cutting an estimated 630,000 tonnes of CO2 emissions annually.



Introduction: India's Largest Metro Network

- DMRC is one of the busiest metro networks in the world, with **10+ color-coded mix gauge lines** across Delhi-NCR.
- **MGT in some of the line is up to 40, annually, with cumulative figures reaching to 615**
- For reliable and robust operation, DMRC follows a preventive maintenance schedule, with scheduled inspections and maintenance of tracks and rolling stock during non-revenue hours (typically post-midnight to early morning).
- For track maintenance, DMRC is employing **preventive rail grinding operation** for both standard and broad gauge lines since **2010** to remove surface defects and for rail reprofiling.





Introduction: Rail Grinding Operation at DMRC

- Currently DMRC has **16-stone Light Rail Grinding (LRG)** Machines, for standard and broad gauge tracks and planning to increase its grinding infrastructure with increasing network length.
- The LRG-18 is designed for dual gauge operation, for both standard and broad gauge tracks.
- The Operations and Maintenance of these machines are carried out by the agency : **M/s Vandhana International**, India.
- DMRC is using different target rail profiles dedicated to high and low rail of the curve to ensure smooth curving operation.
- DMRC has adopted a **proactive frequency based grinding** program at early stages to improvise rail – wheel interaction and asset life extension.





Introduction: Rail Vehicle Stability

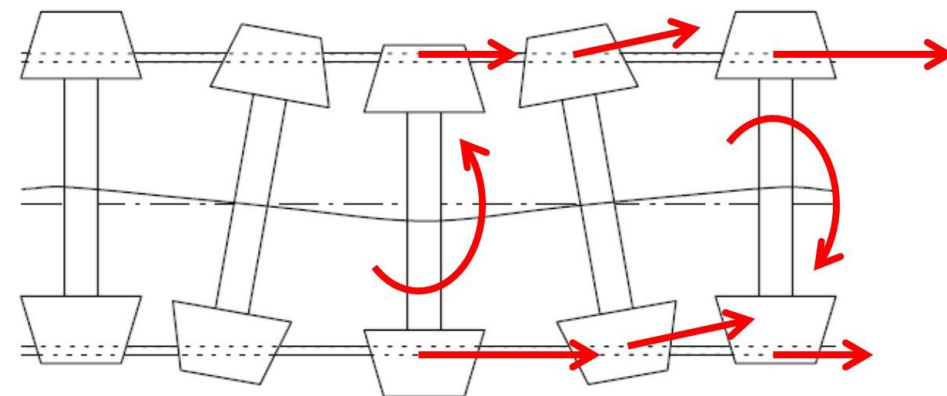
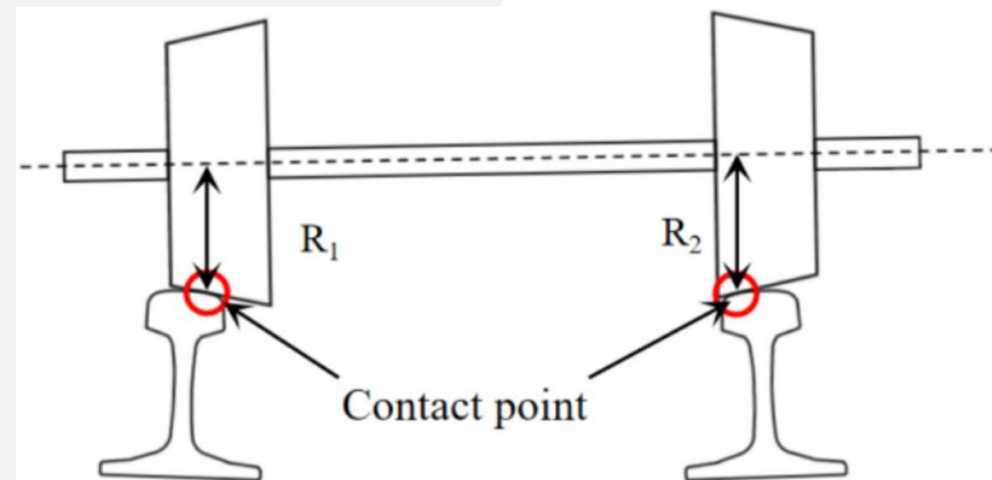
Rail vehicle stability is the ability of a train to move at a given speed without developing growing oscillations in wheel-rail interaction.

Major Factors affecting Rail Vehicle Stability are:

- 1) Wheel-Rail Contact Geometry
- 2) Wheel Profile & Rail Profile Condition
- 3) Track Geometry & Condition
- 4) Operational Parameters

The most dominant factor is **Wheel-Rail Contact Geometry**, represented by:

- 1) Rolling Radius Difference: **Generates restoring/steering forces**
- 2) Equivalent Conicity (λ) : **Primary driver of hunting instability**
- 3) Non-Linearity Parameter: **Characterises unstable dynamic response**
- 4) Contact Patch Characteristics: **Influences creep forces and damping**





Introduction: Effect of Rail and Wheel Profile

- Rail vehicle stability is not a vehicle-only property - it is fundamentally governed by the **evolving wheel - rail contact geometry defined by profiles**.
- Wheel and rail profiles determine the **contact location, rolling radius difference and conicity**, which in turn control creep forces, steering behaviour, and ultimately the stability of the rail vehicle
- **Lateral shift of Wheelset** - Rail & Wheel profile determine how rolling radius changes with lateral displacement
- Worn rail & wheel profile affects the conicity behaviour of vehicle changing the lateral and longitudinal
- Real profiles are **not perfectly conical** Leads to **nonlinear conicity behaviour** resulting **poor dynamic stability**

Benefits of Rail & Wheel Reprofilng

- **Reprofilng restores optimal wheel - rail contact geometry, reducing instability, forces, and wear, thereby improving overall vehicle dynamics.**
- Reprofilng creates a smooth single point wheel - rail contact interface resulting **predictable dynamic behaviour**.
- It reduces and linearise the equivalent conicity which result in **higher critical speed for hunting**.



Introduction: Current Challenges & Practical Insights

▪ Key Challenges in Current Practice

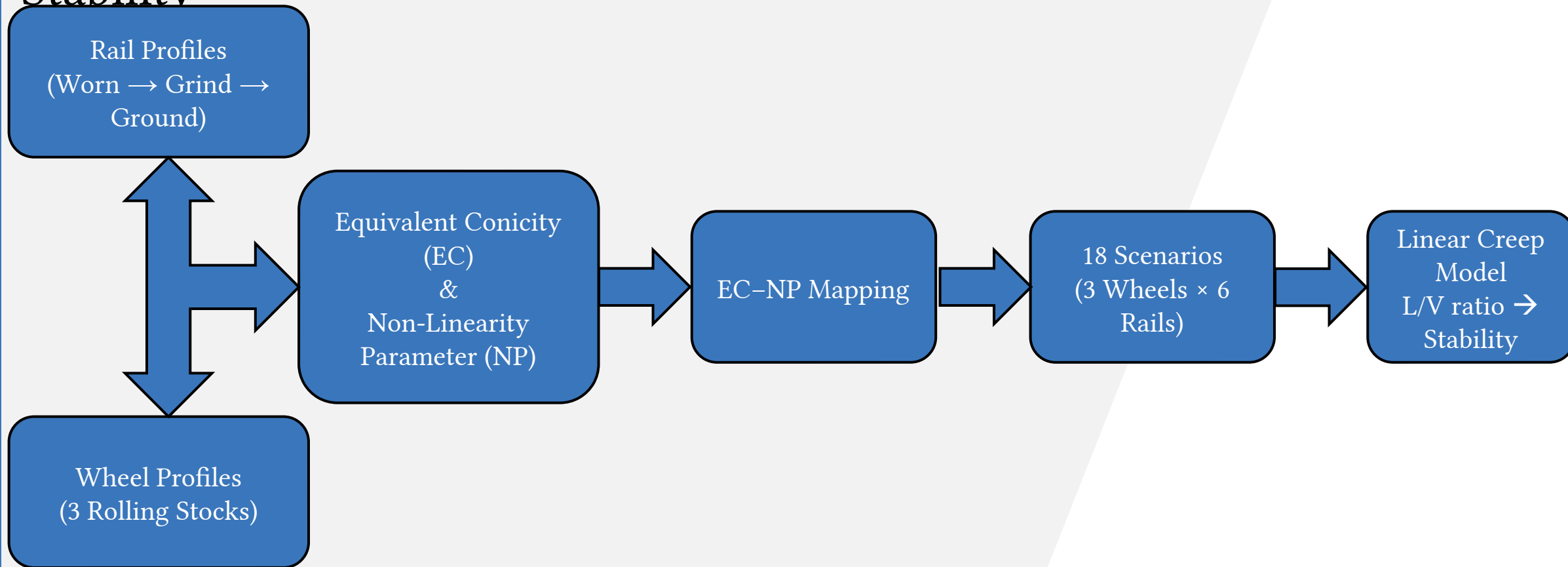
- **Limited field validation in metro systems:** Most conicity and hunting assessments are simulation-driven, with insufficient insights from on-ground observations.
- **Fragmented evaluation of grinding outcomes:** Current practices largely assess conicity in isolation, without jointly evaluating non-linearity (NP) and lateral force (L/V) - critical for understanding actual vehicle dynamic performance.
- **Lack of direct safety benchmarking:** There is minimal field-based correlation of equivalent conicity with L/V behaviour, especially using measured wheel-rail profile pairs.

▪ Value Delivered by This Study

- Provides a **field-driven assessment framework** linking rail/wheel reprofiling to vehicle stability and safety
- Comprehensive dataset: multiple rail and wheel profiles across tangent sections of India's largest metro network for **service as well as ground conditions**
- Establishes a **closed-loop engineering linkage**: Profile condition → Contact geometry (EC, NP) → L/V behaviour → Effect of rail reprofiling
- Enables data-backed decision-making for grinding strategies to improve stability, safety and asset life



Methodology: Framework for Linking Wheel–Rail Profiles to L/V Stability

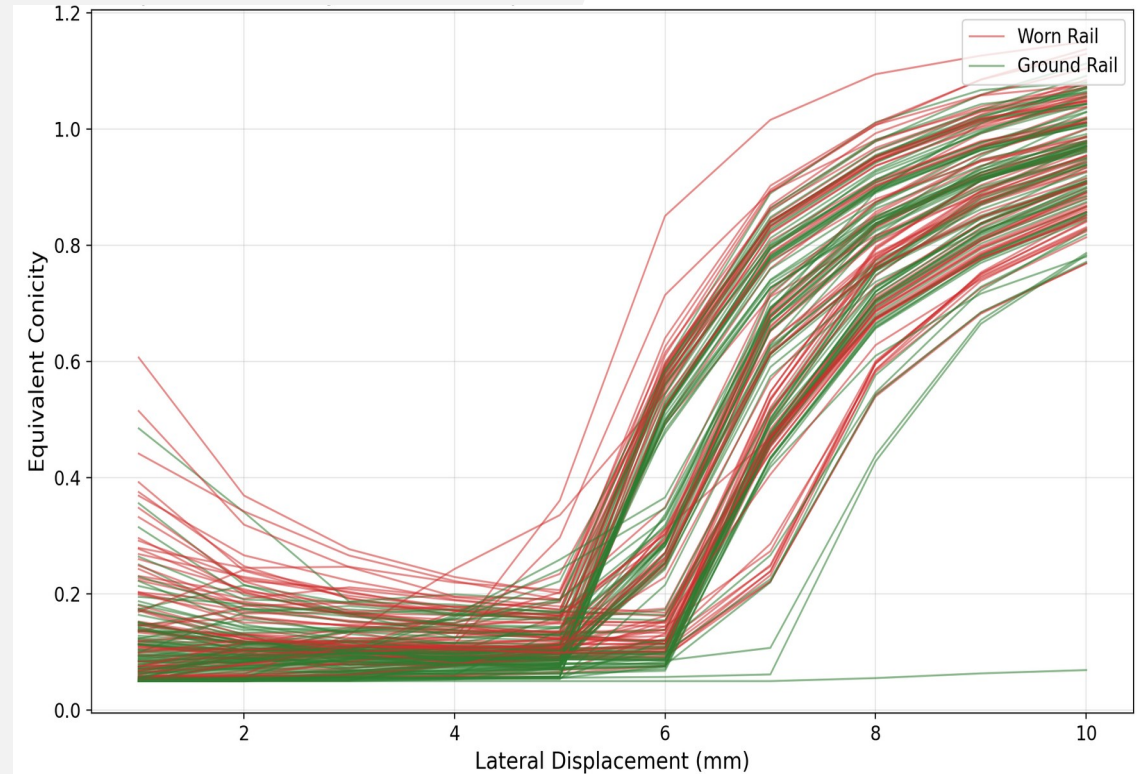
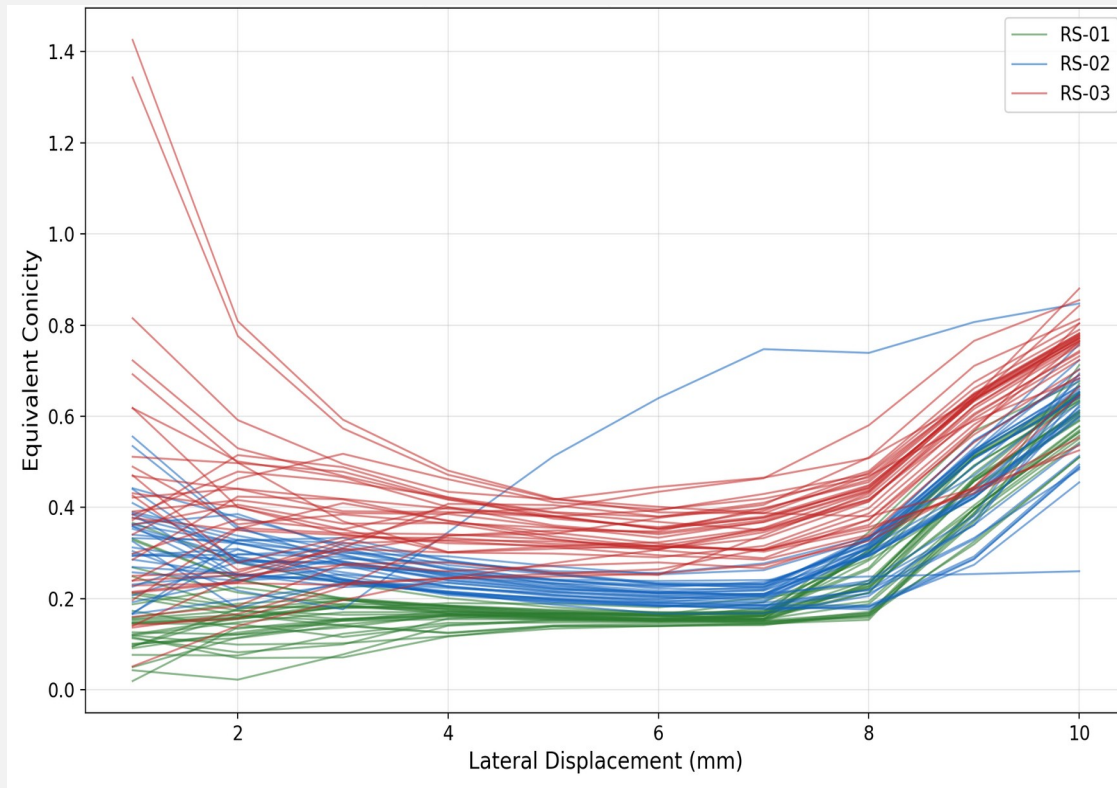


- This dataset captures tangent segments on DMRC broad gauge lines, with rail profile measurements taken repeatedly **before and after rail grinding operations**.
- Across rolling stocks with multiple wheels assessed at varying operational days since their last reprofiling.
- The complete data set provides a **wide range of in-service rail and wheel conditions**.



Result: Rail & Wheel Conicity Progression

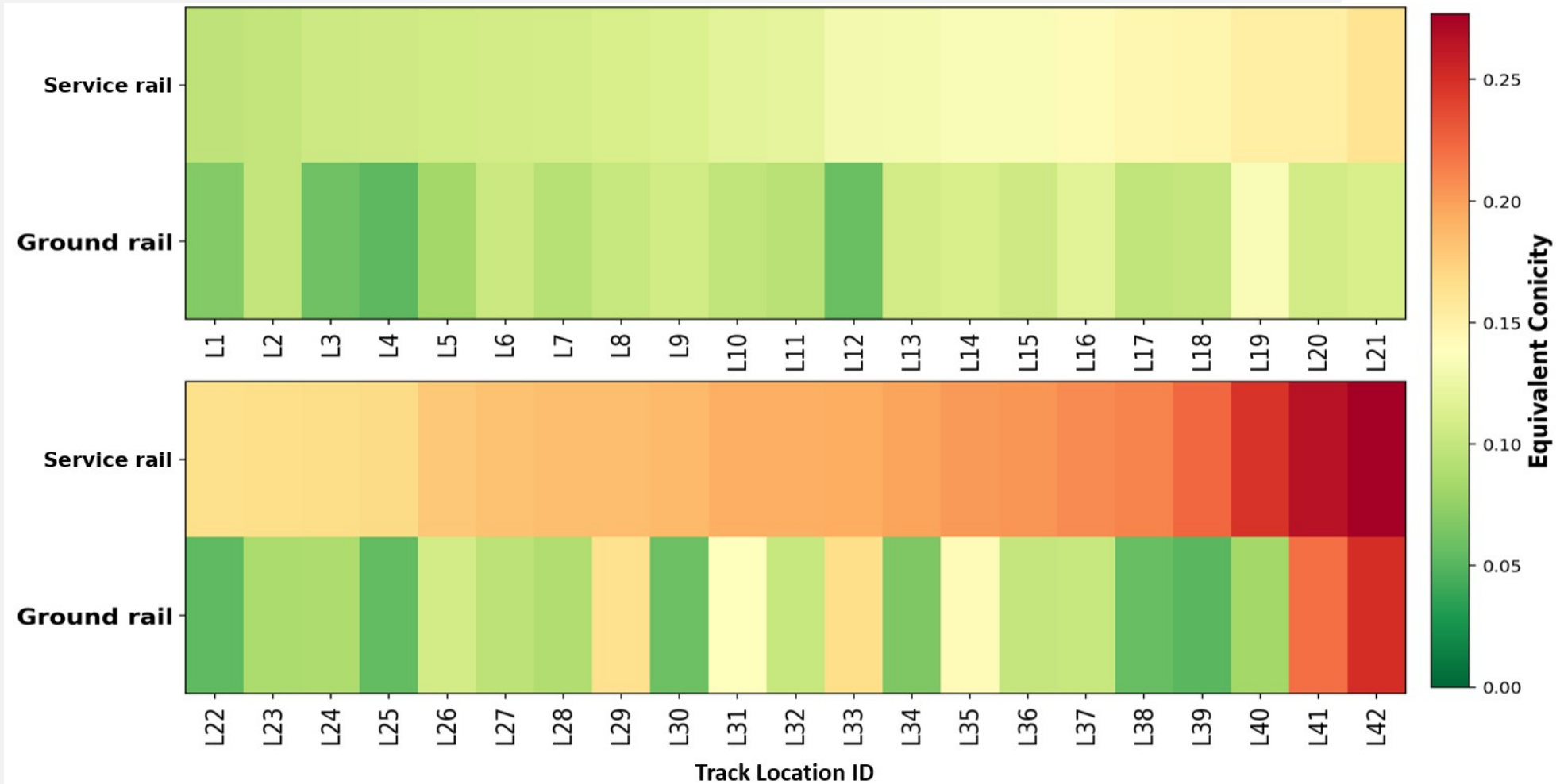
- Conicity progression of rail and wheel is analysed with respect to standard wheel (SK-91 146) and rail (UIC-60) profiles respectively.
- With increase in operational days, the conicity for wheel rises mainly due to thread wear.
- Ground rail shows **consistently improved conicity** behaviour than Worn on both lines at lower displacement





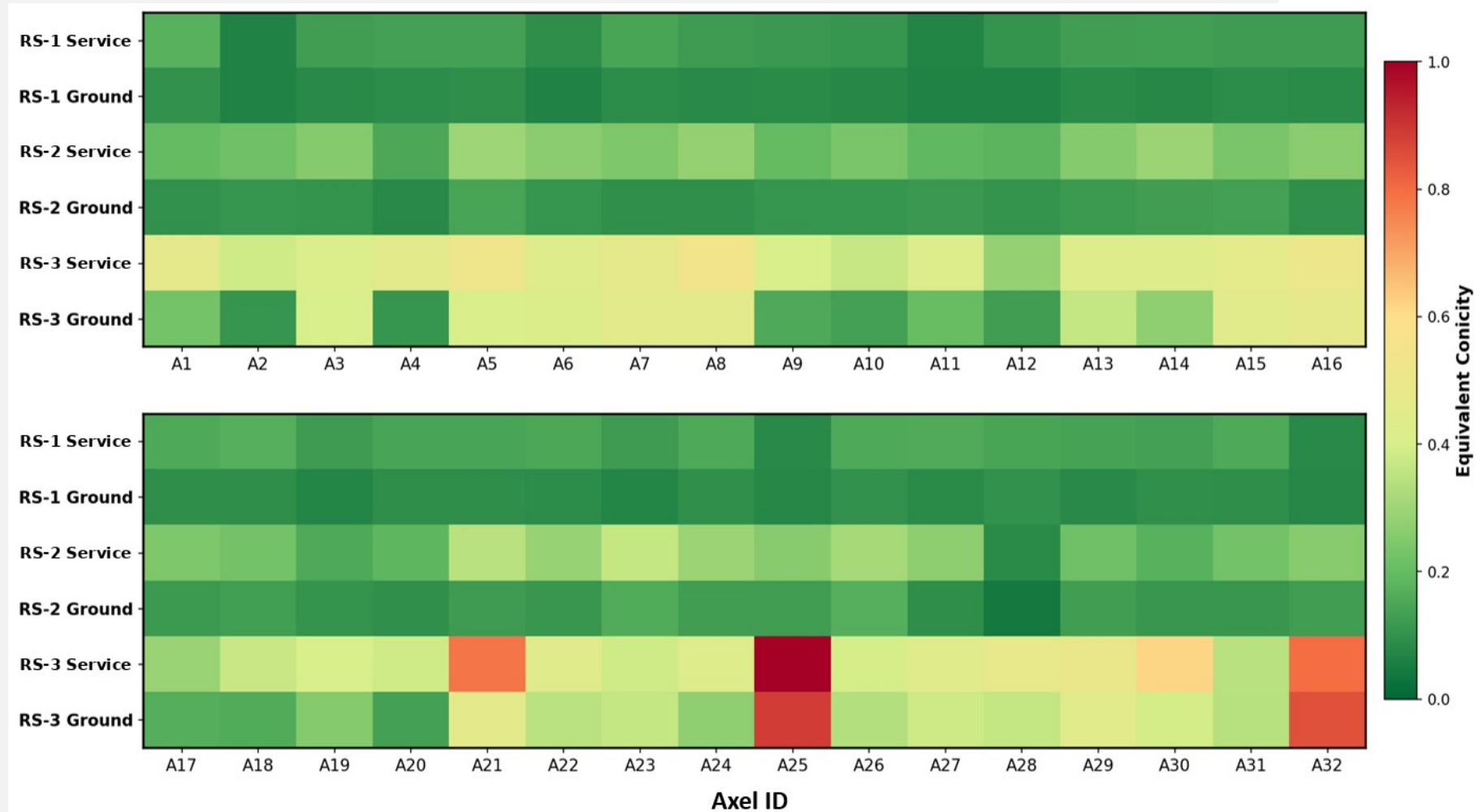
Result: Grinding Reduces & Stabilizes Conicity

- Rail grinding uniformly improves the **conicity behaviour across all multiple locations**.
- Track Locations are arranged with respect to increasing conicity of worn rail.



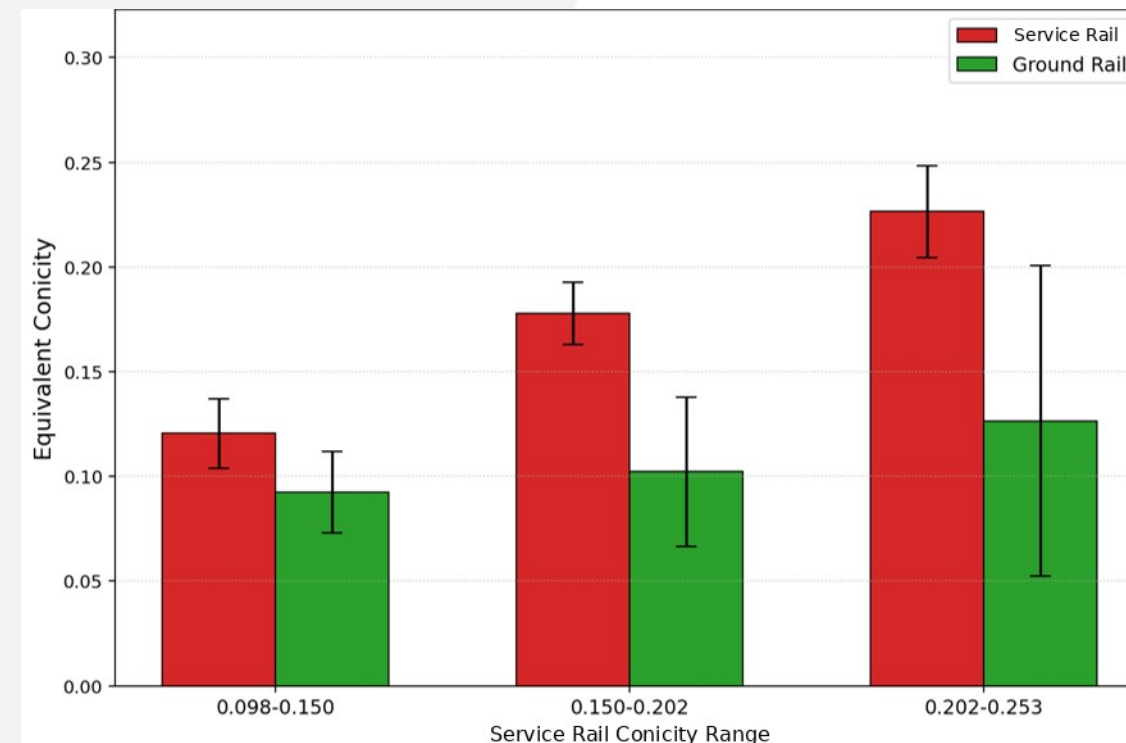
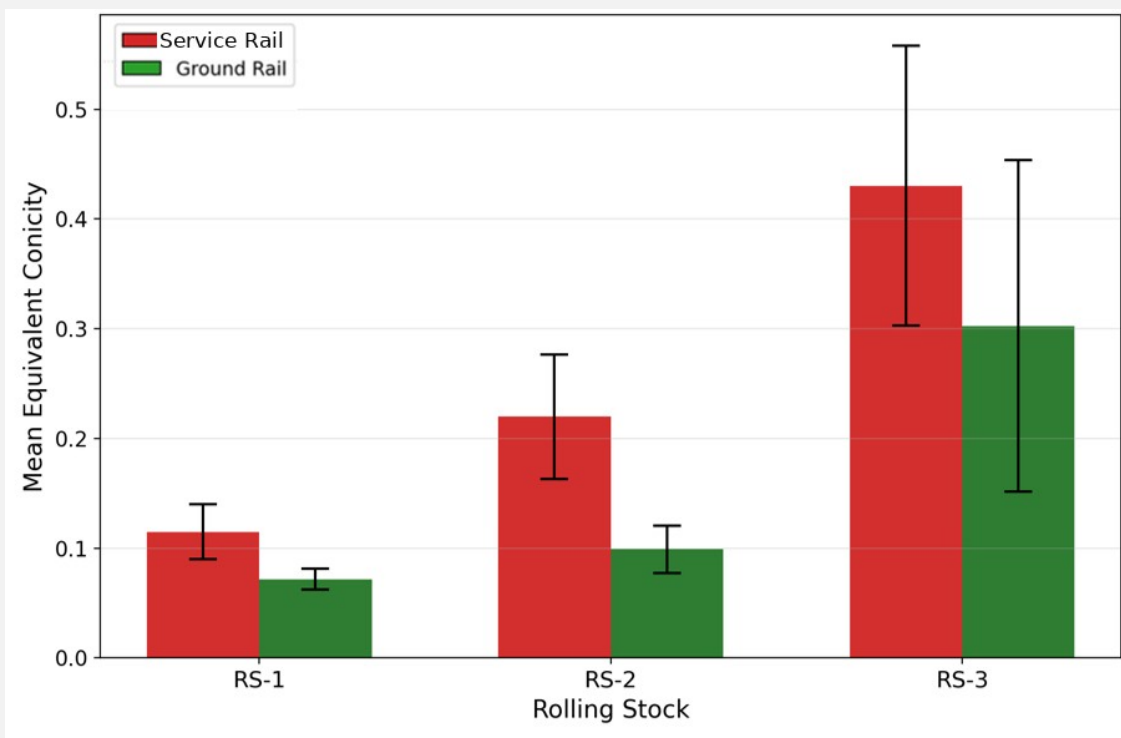
Result: Rail Grinding Effectiveness Varies by Rolling Stock

- **Grinding consistently reduces conicity** across all rolling stocks — even for the most severely worn wheels.
- Rail grinding delivers a meaningful improvement, though there is lesser improvement in worn wheels.



Result: Rail Grinding Consistently Reduces Conicity

- Grinding delivers a consistent, substantial conicity improvement across the fleet's rolling stock (**RS-1: ~37%; RS-2: ~55%; RS-3: ~30%**)
- Grinding achieve uniform conicity** for all worn rail conditions.
- Grinding reduces mean conicity by **23% in the low range**, and by **~42–44% in the medium and high worn-conicity ranges**.



Result: Non-Linearity of Conicity Profile

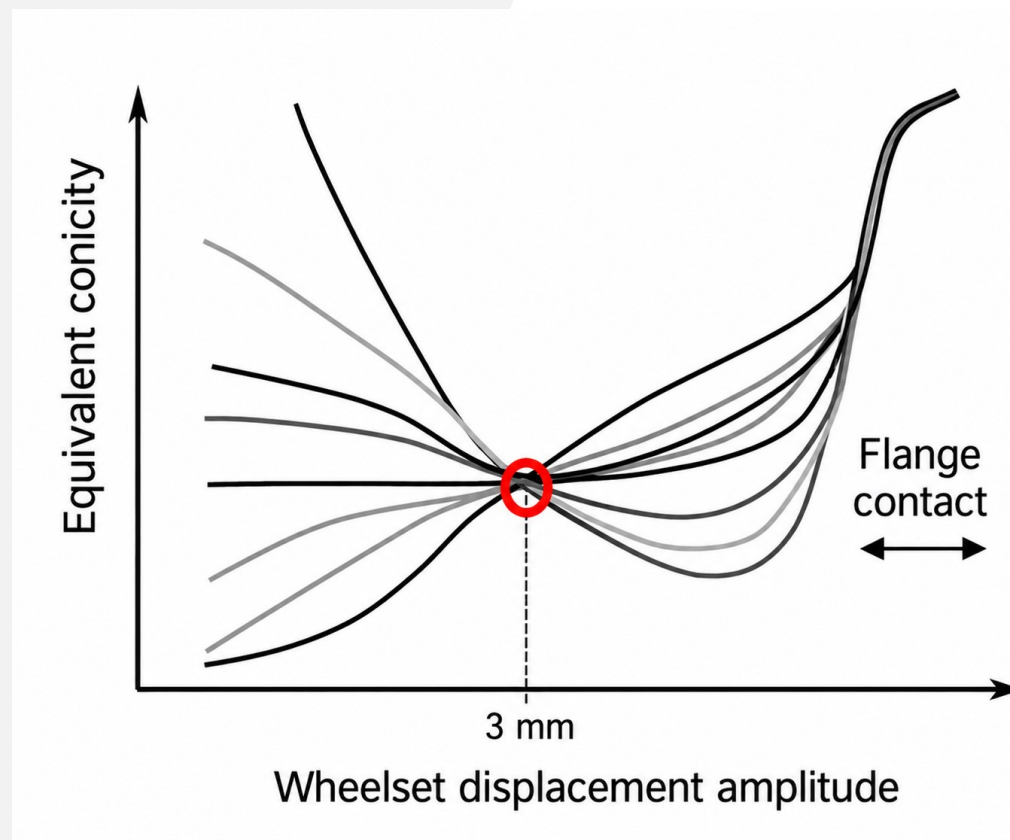
- Two wheel-rail pairs can have identical conicity value at a point but completely different conicity behavior with lateral displacement.
- The Nonlinearity Parameter (NP) quantifies the shape of the equivalent conicity function — specifically, how the conicity changes as function of wheelset's lateral displacement.

$$NP = \frac{\lambda_{4mm} - \lambda_{2mm}}{2}$$

Here, λ_{4mm} and λ_{2mm} are the conicity values at 4 mm and 2 mm of wheelset lateral displacement respectively

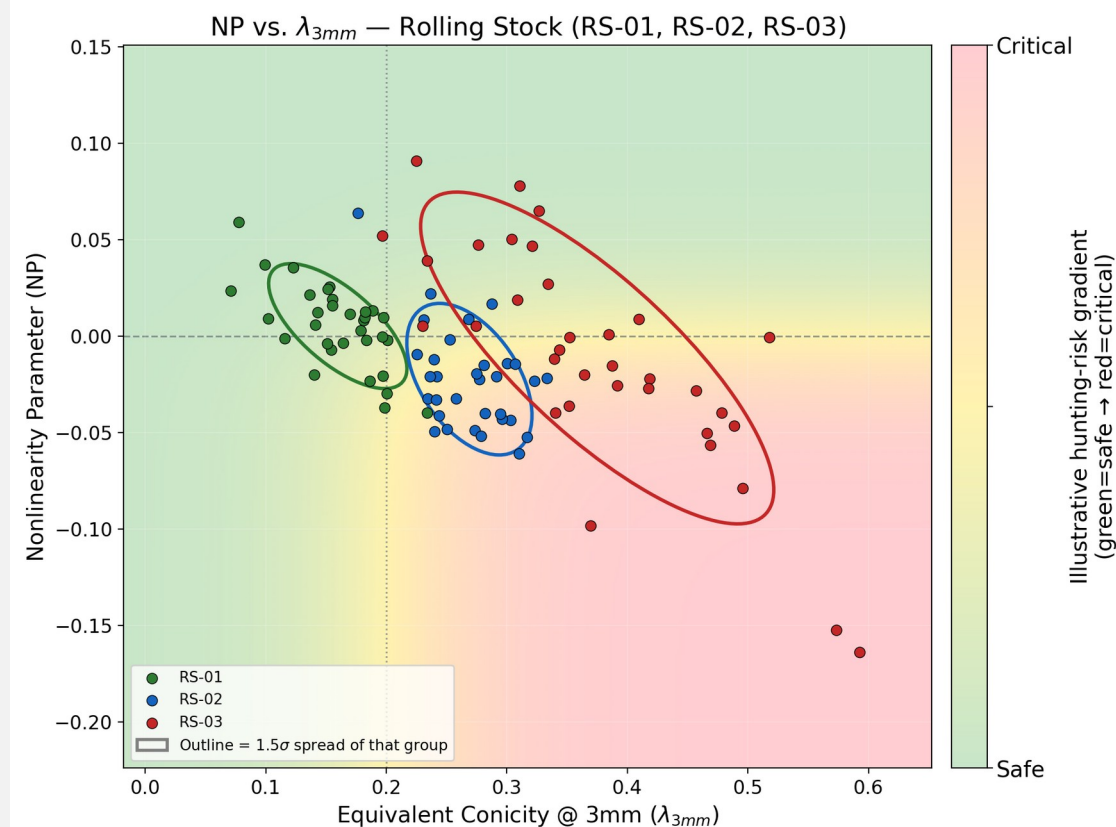
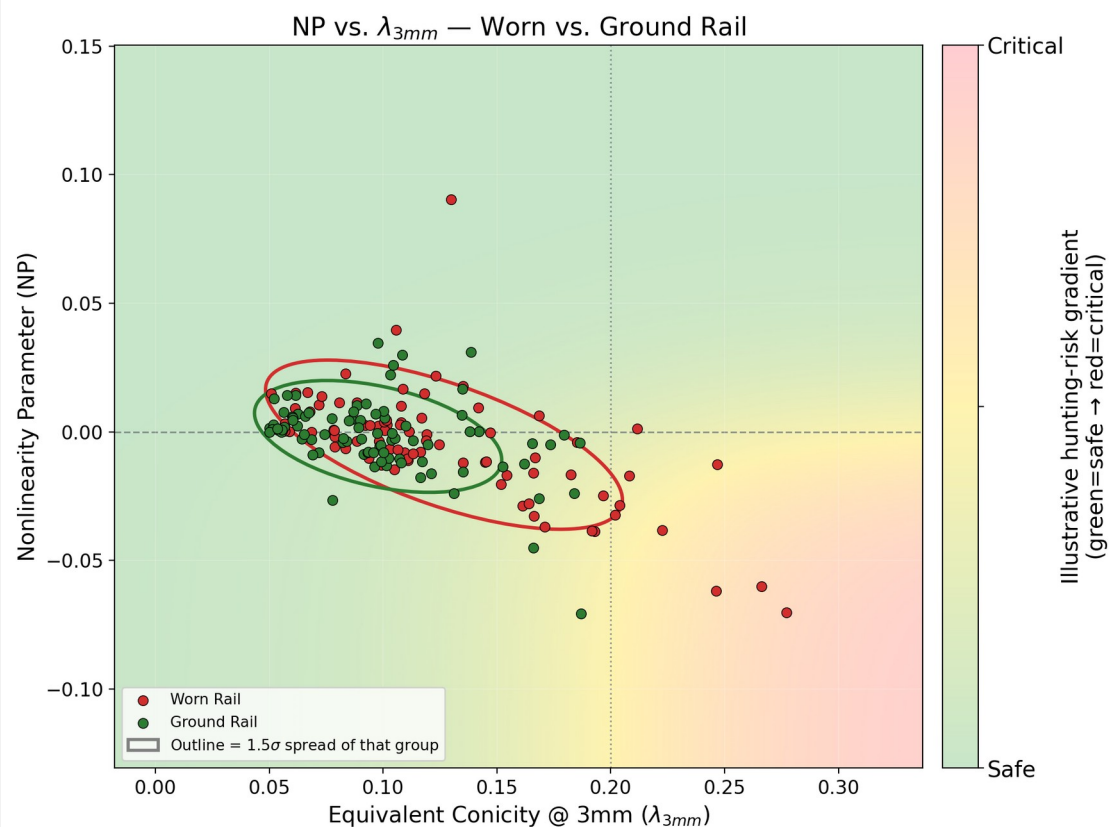
Condition:

- NP > 0 (Type A): The forgiving case – conicity rises smoothly with displacement – hunting if occurs, needs a large disturbance to trigger, and decays smoothly as speed drops.
- NP < 0 (Type B): The dangerous case – conicity is high early and flattens, contact jumps abruptly toward the gauge corner – no trigger threshold needed for hunting to occur



Result: Hunting Risk Zone Mapping – NP vs. Conicity

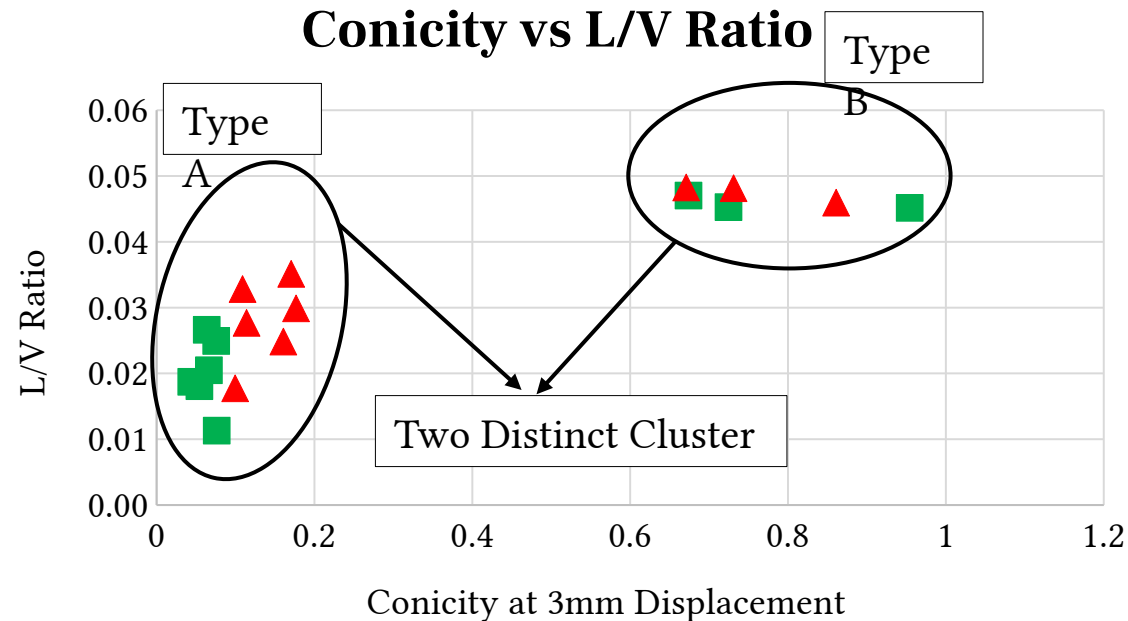
- NP vs λ_{3mm} plots for rail (worn and ground) and different rolling stocks are shown below.
- Both plots confirm the same inverse trend.
- Rail & wheel wear both affects the hunting risk, conicity for wheel rises with time in service
- Grinding mitigates the risk** – Worn's ellipse stretches further into the high- λ /negative-NP tail than ground's.





Result: L/V variation with Conicity

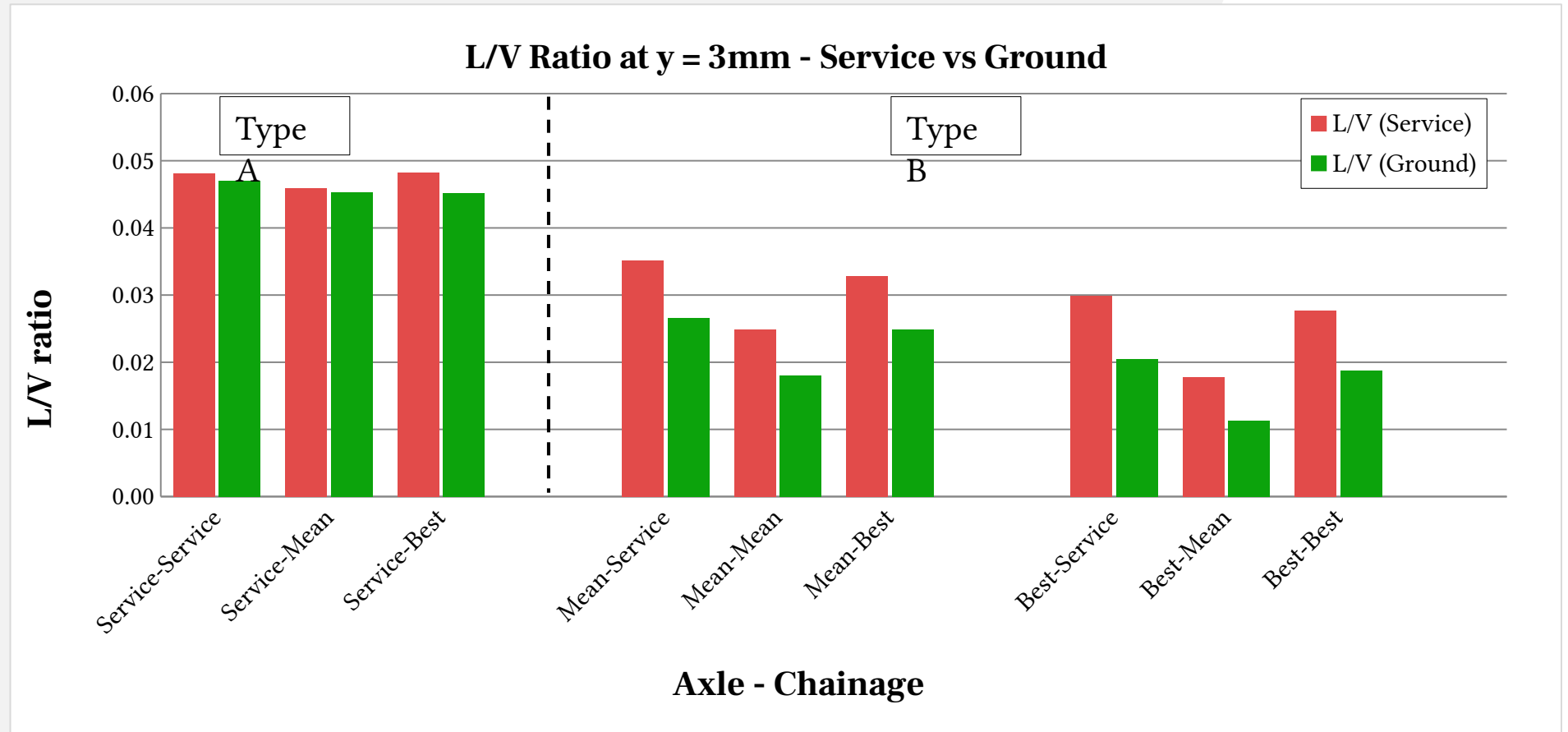
- **Different combination of wheel and rail profiles** were analysed.
- **Two distinct clusters**, not a continuous trend - low-conicity points sit clearly separated from high-conicity points.
- grinding providing L/V benefit within conicity below 0.2, whereas grinding's benefit reduces in high conicity.
- Type A cluster, **L/V rises with conicity proportionally**
 - Higher conicity → higher spin creepage → higher creep force → higher L/V.
- For Type B cluster, **force is capped by friction, not geometry**
 - Rail reprofiling does not change L/V longer significantly.





Result: L/V Values for Different Condition

- **Grinding is effective at low displacement across all combinations.**
- **Type B** combinations show the largest grinding benefit (**20–30% L/V reduction**), attributable to improved contact conformity following rail reprofiling.
- **Type A** combinations show a smaller, but still measurable, benefit (**2–6%**).





Key Findings: Timely Asset Reprofilng Helps in Vehicle Stability

- **Conicity rises with time in service:** Rail operation result in change in wheel-rail contact geometry over time due to rail and wheel wear. Reprofilng of these assets on time helps to maintain predictable vehicle dynamic behaviour.
- **Non-linear behaviour of conicity:** The hunting susceptibility does not only depends on single conicity value at different lateral displacement. It also depends the extent of non-linearity of conicity curve with lateral displacement.
- **Asset Reprofilng effect on Hunting:** Rail and wheel reprofilng consistently result in longer hunting wavelengths, which leads to smoother and safe operating condition for tangent tracks.
- **Two clusters of L/V vs Conicity:** For type A cluster, L/V varies proportionally with conicity change, directly impacting creep forces. However, for type B cluster, forces are controlled by friction not by rail wheel geometry due to sever wheel and rail wear.
- **Effect of rail profile:** Although the rolling-stock has greater impact on vehicle stability but even for same axel conditions rail reprofilng exhibited reduction in conicity, transition of non-linearity parameter towards safety and also significant reduction in L/V characteristics.



Path Forward: Towards data-driven vehicle stability management

- **Screening indicators to full dynamic validation:** This study uses linear creep theory + Shen-Hedrick-Elkins saturation (a hand-calculable proxy) plus static indicators (NP, hunting wavelength) as a screening layer. The natural next step is to include track irregularities to predict the dynamic responses.
- **Cross-sectional to longitudinal tracking.** Current study compares cross-section across different wheels and rail. Following the same rail – wheel combination sets across multiple reprofiling cycles would show the actual conicity growth rate.
- **Extend beyond tangent track:** The study is explicitly scoped to tangent sections. Evaluation of dynamic characteristics of Rail – wheel interaction during curving will provide a complete understanding of vehicle stability for DMRC broad gauge tracks.



Thank You

Safety Through Stability Starts With Matching Maintenance Strategy To Actual Rail & Wheel Condition.